User Manual

KALI-LED-IPED-NUDETECTIVE

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1. Description

The bootable ISO KALI-LED-IPED-NUDETECTIVE is a customization of the Kali Linux forensic distribution built to include the LED, IPED, and NuDetective applications, aiming to execute them at search and seizure locations.

The advantage of this approach is practicality: at the search location, the examiner can simply configure the target machine to boot from the USB port, loading the Linux operating system. Upon initialization, the forensic mode boot option must be selected (executed by default) to avoid automatically mounting swap and partitions.

On the desktop, there are links for the automatic execution of LED, "IPED Launcher", and NuDetective on the detected file systems or block devices (IPED hard drives).

2. Boot Instructions

The main care when using this tool is when configuring the examined machine to boot from the USB drive.

2.1 BIOS/UEFI Shortcut Keys

Each manufacturer usually uses a different key to enable the boot drive selection. Some common keys are **ESC, F1, F2, F10, F12, DEL**.

Computer Model	BIOS Key/s	Boot menu (UEFI)
Acer	F1, F2, Ctrl+Alt+Esc	Esc F2,F9 ,F12
Asus	–	Esc, or F8
AST	Ctrl+Alt+Del, Ctrl+Alt+Esc	–
Dell (All models)	–	F12
HP	–	ESC or F9
Lenovo (all models)	–	F12
Panasonic CF-25	F1	–
Samsung (all models)	–	F12
Sony Vaio (All Models)	–	F11
Toshiba (All Models)	–	F12

Computer Model	BIOS Keys
Acer®	F1, F2, CTRL+ALT+ESC
AST®	CTRL+ALT+ESC, CTRL+ALT+DEL
Compaq® 8700	F10
CompUSA®	DEL
Cybermax®	ESC
Dell® 400	F3, F1
Dell Dimension®	F2 or DEL
Dell Inspiron®	F2
Dell Latitude	Fn+F1 (while booted)
Dell Latitude	F2 (on boot)
Dell Optiplex	DEL
Dell Optiplex	F2
Dell Precision™	F2
eMachine™	DEL
Gateway® 2000 1440	F1
Gateway 2000 Solo™	F2
HP®(Hewlett-Packard)	F1, F2
IBM®	F1
IBM E-pro Laptop	F2
IBM PS/2®	CTRL+ALT+INS after CTRL+ALT+DEL
IBM Thinkpad® (newer)	Start | Programs | Thinkpad CFG
Intel® Tangent	DEL
Micron™	F1, F2, or DEL
Packard Bell®	F1, F2, DEL
Sony® VIAO	F2, F2
Tiger	DEL
Toshiba® 335 CDS	ESC
Toshiba Protege	ESC
Toshiba Satellite 205 CDS	F1
Toshiba Tecra	F1 or ESC

2.2 Configuration on Modern Machines (Windows 10/11 and UEFI)

On newer machines, running Windows 10 and above with UEFI, the Secure Boot option must be disabled. To do this, in some cases, it is necessary to boot into Windows, select restart, and click the restart button while holding the Shift key.

Restarting this way will open a recovery interface, where you should navigate to **"Troubleshoot" -> "Advanced Options" -> "UEFI Firmware Settings"**. The computer will restart into the UEFI configuration interface, where you can disable Secure Boot. This option can be re-enabled after the analysis is complete.

.2.3 Technical Boot Notes

- **Forensic Mode:** It is always recommended to select the "**Live (forensic mode)**" option at startup (pre-selected to start in 5s), to prevent the automatic mounting of disks and swap partitions. If there are compatibility issues, you can try the "Live (amd64 failsafe)" option**.**

- **Disk Visibility:** On some computers, the internal disk may not be visible in Linux. In this case, ensure that the SATA option in the BIOS is set to **AHCI**, and not RAID.

2.4 Version with NVIDIA and AMD drivers (GPU Acceleration)

There is a specific version of the ISO that includes proprietary **NVIDIA (CUDA)** and **AMD (ROCm)** drivers. This version is optimized to take advantage of GPU processing for detecting child sexual abuse via AI (PyTorch) within IPED.

- **Universal Compatibility:** Although focused on hardware acceleration, this version works perfectly on systems that only have a CPU (without a dedicated graphics card).
- **Automatic Optimization:** IPED detects the presence of compatible hardware and enables GPU acceleration automatically, drastically reducing processing time.
- **PyTorch Environment:** If you want to use PyTorch functionalities in the Python environment via the command line, simply execute the appropriate command:
    - `source /opt/venv-cuda/bin/activate`
    - `source /opt/venv-cuda-legacy/bin/activate`
    - `source /opt/venv-cuda-rocm/bin/activate`

3. Mounting and Decryption Features

This distribution includes scripts that automatically allow the decryption of disks with file formats supported by Kali Linux, as well as Windows RAID partitions, BitLocker encrypted disks, and the mounting of Windows Virtual Shadow Copy (VSS) partitions.

3.1 VSS - Windows Virtual Shadow Copie

Windows Virtual Shadow Copy drives can be mounted using the "**Mount Windows VSS**" script on the desktop..

Windows automatically generates these backups at specific times (e.g., system updates). They contain copies of all files on the volume, making them useful for recovering deleted files.

- **Time Warning:** Each VSC has a complete disk snapshot. The processing time by applications will be multiplied by the number of VSCs found. Use with caution.

3.2 BitLocker Usage

Kali natively recognizes BitLocker partitions (indicated by an icon with a red circle and the description "Encrypted").

- **Suspended Encryption:** If suspended via Windows (Manage BitLocker -> Suspend Protection), the scripts will automatically mount the partition as read-only. **Caution**: when rebooting into Windows, the suspend mode is terminated.
- **Recovery Key: If the protection is not suspended, the 48-digit key will be requested when executing the applications.**

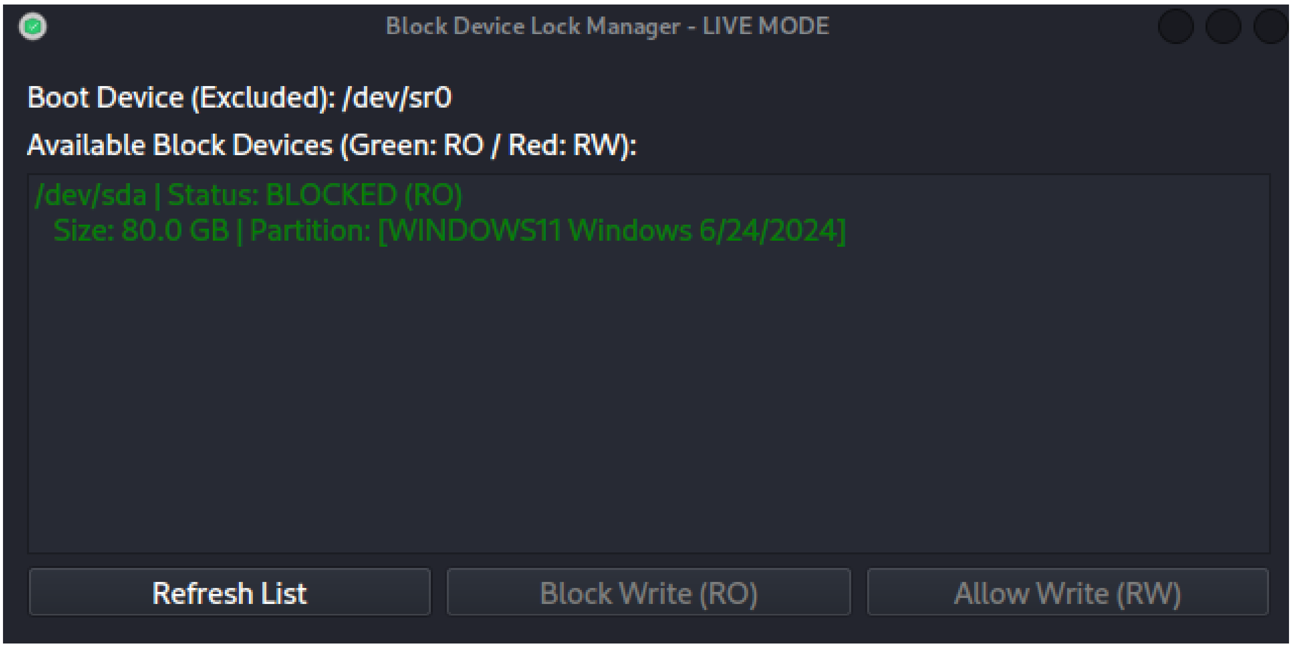
4. Tool Operation

4.1 Write Blocking (Device Level)

ATTENTION: At startup, all disks are configured as read-only at the block device level, using the `blockdev --setro`. Even if you click on the desktop icons, they will be mounted as RO (Read-Only).

To enable writing (e.g., to save data to an external flash drive):

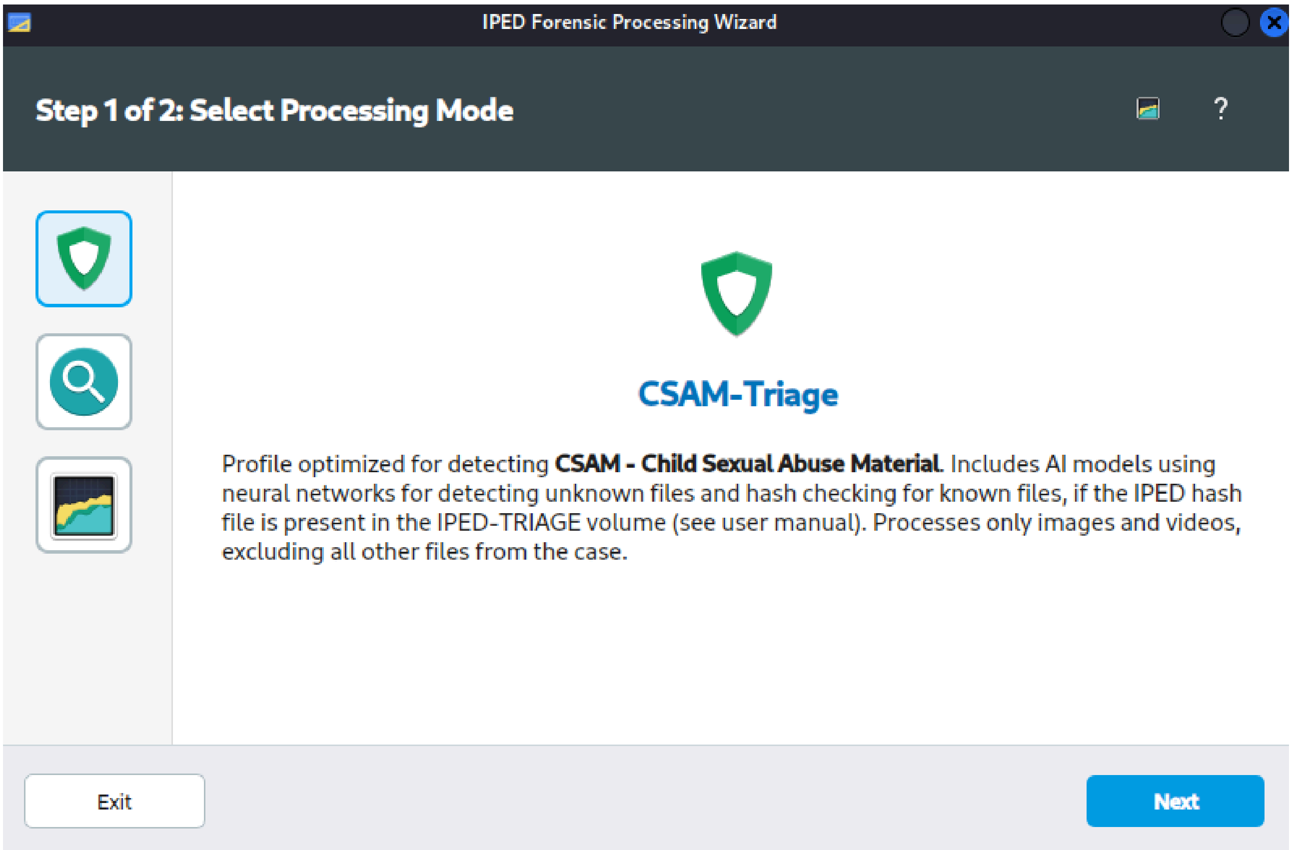
1. Execute the "**Disk Shield**" application.
2. The device icon will turn **red**, indicating that writing has been enabled.



4.2 Applications

IPED Launcher

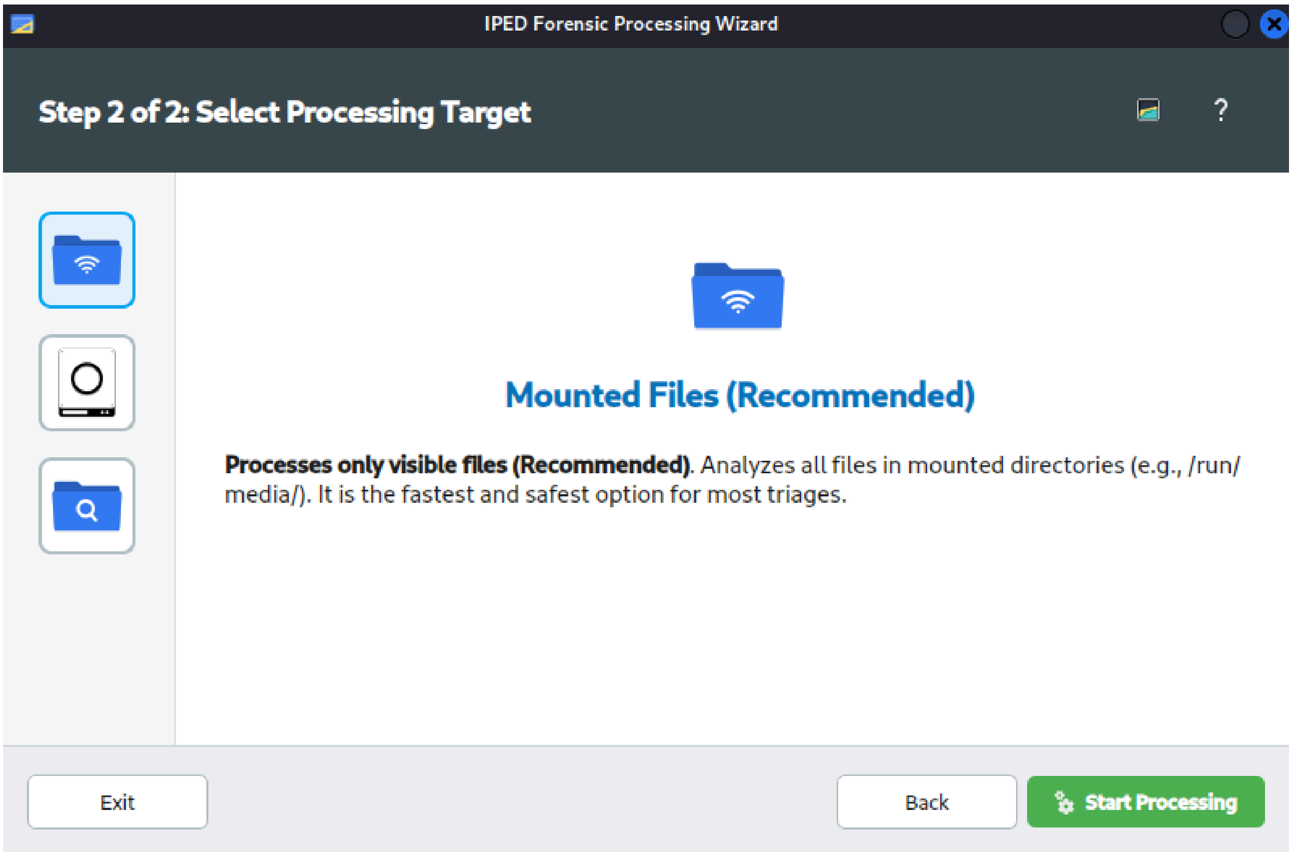
A graphical interface that allows configuring advanced forensic processing and triage profiles using IPED.



Perfis:

- **CSAM Triage:** Optimized for child abuse material detection. Uses AI models (neural networks) for detecting unknown files and hash verification for known files (if the IPED hashes database is present in the IPED-TRIAGE partition). Processes only images and videos, excluding all other case files.
- **Triage:** Indexes documents (Office, PDF, emails, history). Media parsers are disabled. Useful for textual searches in triage scenarios.
- **Fastmode:** Ultra-fast mode. Runs an "ls" (list) command on the file tree. Allows navigating the gallery and viewing metadata (MAC times) without calculating hashes or indexing content.

Processing Targets:



- **Mounted Files (Recommended):** Processes only visible files. Analyzes all files in mounted directories (e.g., /run/media/). It is the fastest and safest option for most triages.
- **Disks (Full/Slow):** Processes physical devices, LDM (RAID), VSS, and BitLocker directly.
- **Select Directory/Image:** Allows manually choosing a specific target (e.g., .E01, .ufdr, directory).

5. IPED-TRIAGE Partition Configuration

For the "Triage" and "CSAM Triage" profiles, execution in RAM may fail due to a lack of memory. It is recommended to create an extra partition on the USB/SSD disk used for booting.

If using Ventoy, simply rename the exFAT partition from "Ventoy" to "IPED-TRIAGE", or create an additional exFAT partition named so on the disk.

Procedure with Balena Etcher:

1. After burning the ISO to the USB disk, open Windows Disk Management (diskmgmt.msc), click Cancel if it asks to initialize the disk, and press F5 to refresh the list of disks and partitions.
2. Create a new partition in the remaining free space of the disk where the ISO was burned, using the specifications from the table below.

Partition Name	`IPED-TRIAGE` (All uppercase)
Format	**exFAT** (preferred)
Hashes Database	`/IPED-HASHESDB/iped-hashes.db`
Keywords	`/palavras-chave.txt` (optional)

5.1 Virtual Memory Usage (Swap)

For machines with less than 4GB of RAM, use the "Virtual Memory" (Create Virtual Memory) script. If the IPED-TRIAGE drive in exFAT format is detected, it will create a swapfile to prevent system freezes. In IPED's triage and csam_triage profiles, this file is created automatically if the IPED-TRIAGE partition is available.

6. Disk Generation and Hardware

Flash Drive Preparation

- Minimum 8GB capacity.
- The use of Balena Etcher is recommended, although Ventoy is also supported.

Hardware Suggestion

It is recommended to burn the ISO to an **external SSD via USB 3.0**. This significantly reduces processing time compared to standard USB devices, in addition to providing higher random read speeds, which are necessary for the efficient use of virtual memory.

7. Contact and Support

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